UDLAP

Homework No.3: Key-Based Enhancement of DES and the Enigma Machine.

OBJECTIVE

In this homework, students will explore an enhanced version of the Data Encryption Standard (DES) algorithm [1], referred to as KE-DES (Key-based Enhanced DES) [2]. Although KE-DES is not cryptographically secure, its operations are quite similar to those of the original DES algorithm. To complete this task, students should read the description of KE-DES available on the Blackboard website.

PROCEDURE

Please follow the steps outlined in the KE-DES scheme to implement it in any programming language. Create your own KE-DES function rather than using any pre-existing functions. Your results should focus on two different 64-bit datasets and must demonstrate the KE-DES process, including key generation time, KE-DES encryption, and KE-DES decryption. Additionally, investigate a brute-force attack along with cryptanalysis methods, specifically differential and linear attacks on KE-DES, and include these findings in your report. Lastly, discuss the relationship between KE-DES and the DES algorithm.

An investigation of the **Enigma machine**, including the processes of encryption and decryption as well as cryptanalysis, should also be provided. Include in this report (as an appendix) the computed addition and multiplication tables for the $\mathrm{GF}(2^3)$ (see PP slide 124).

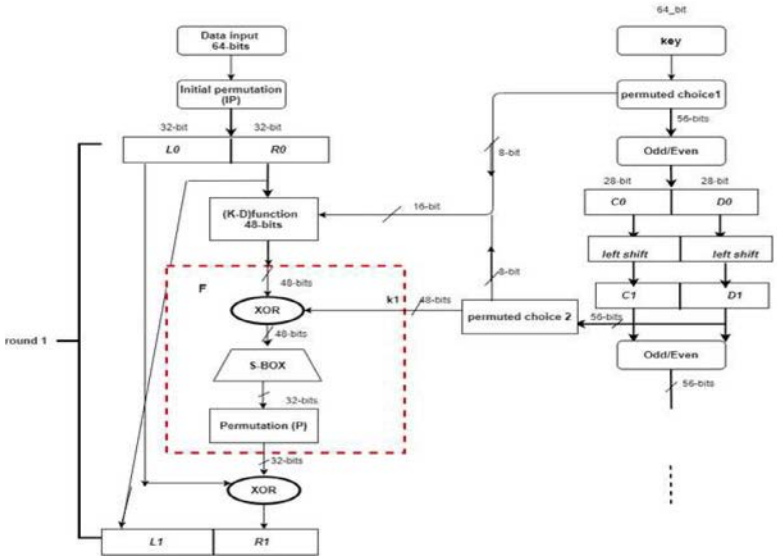


Figure 1: Architecture of Key-Based Enhancement of Data Encryption Standard (see [2]).

REFERENCES

[1]. William Stallings, *Cryptography and Network Security: Principles and Practices,* 7th Edition, Pearson Prentice Hall, 2016.

[2]. O. Reyad, H. M. Mansour, M. Heshmat, and E. A. Zanaty, Key-Based *Enhancement of Data Encryption Standard for Text Security*, IEEE National Computing College Conference, 2021.

DUE DATE: WEDNESDAY, 14 OCTOBER 2026